

DBMS Assignment 2

ER Diagram Design and Relational Schema

Application / Domain: Online Food Delivery System

This document builds on the tables identified in DBMS Assignment 1 (Online Food Delivery System) and develops them into a complete Entity–Relationship (ER) model and a normalized relational schema. All eight entities from Assignment 1 — Customer, Address, Restaurant, Menu_Item, Delivery_Partner, Orders, Order_Item and Payment — are retained, so the ER model matches the application scope selected earlier.

1. Entities and Attributes

Each entity below lists its attributes, data types, and its Primary Key (PK). Foreign Keys (FK), where present, are shown in purple and are explained fully in Section 4.

1.1 Customer

Attribute	Data Type	Key	Description
Customer_ID	INT	PK	Unique identifier for each customer
Name	VARCHAR(100)		Full name of the customer
Email	VARCHAR(150)	UNIQUE	Login email address
Phone	VARCHAR(15)	UNIQUE	Primary contact number
Password	VARCHAR(255)		Hashed account password

1.2 Address

Attribute	Data Type	Key	Description
Address_ID	INT	PK	Unique identifier for each saved address
Customer_ID	INT	FK	Owning customer (references Customer)
House_No	VARCHAR(30)		House / flat / apartment number
Street	VARCHAR(150)		Street name
City	VARCHAR(100)		City name
Pincode	VARCHAR(10)		Postal code

1.3 Restaurant

Attribute	Data Type	Key	Description
Restaurant_ID	INT	PK	Unique identifier for each restaurant
Restaurant_Name	VARCHAR(150)		Name of the restaurant
Phone	VARCHAR(15)		Restaurant contact number
Address	VARCHAR(255)		Physical location of the restaurant
Rating	DECIMAL(2,1)		Average customer rating

1.4 Menu_Item

Attribute	Data Type	Key	Description
Item_ID	INT	PK	Unique identifier for each menu item
Restaurant_ID	INT	FK	Restaurant offering the item (references Restaurant)
Item_Name	VARCHAR(150)		Name of the food item
Description	VARCHAR(500)		Brief description of the item
Price	DECIMAL(10,2)		Current unit price
Availability	BOOLEAN		Whether the item is currently orderable

1.5 Delivery_Partner

Attribute	Data Type	Key	Description
Delivery_ID	INT	PK	Unique identifier for each delivery partner
Name	VARCHAR(100)		Full name of the delivery partner
Phone	VARCHAR(15)	UNIQUE	Contact number
Vehicle_Number	VARCHAR(20)	UNIQUE	Registered vehicle number
Status	VARCHAR(30)		Current availability, e.g. Available / On Delivery

1.6 Orders

Attribute	Data Type	Key	Description
Order_ID	INT	PK	Unique identifier for each order
Customer_ID	INT	FK	Customer who placed the order (references Customer)
Restaurant_ID	INT	FK	Restaurant fulfilling the order (references Restaurant)
Address_ID	INT	FK	Delivery address for the order (references Address)
Delivery_ID	INT	FK	Delivery partner assigned (references Delivery_Partner)
Order_Date	DATETIME		Timestamp when the order was placed
Order_Status	VARCHAR(30)		e.g. Placed, Preparing, Out for Delivery, Delivered, Cancelled
Total_Amount	DECIMAL(10,2)		Total value of the order

1.7 Order_Item (associative entity)

Attribute	Data Type	Key	Description
Order_ID	INT	PK, FK	References Orders
Item_ID	INT	PK, FK	References Menu_Item
Quantity	INT		Number of units of the item ordered
Unit_Price	DECIMAL(10,2)		Price of the item at the time of ordering

1.8 Payment

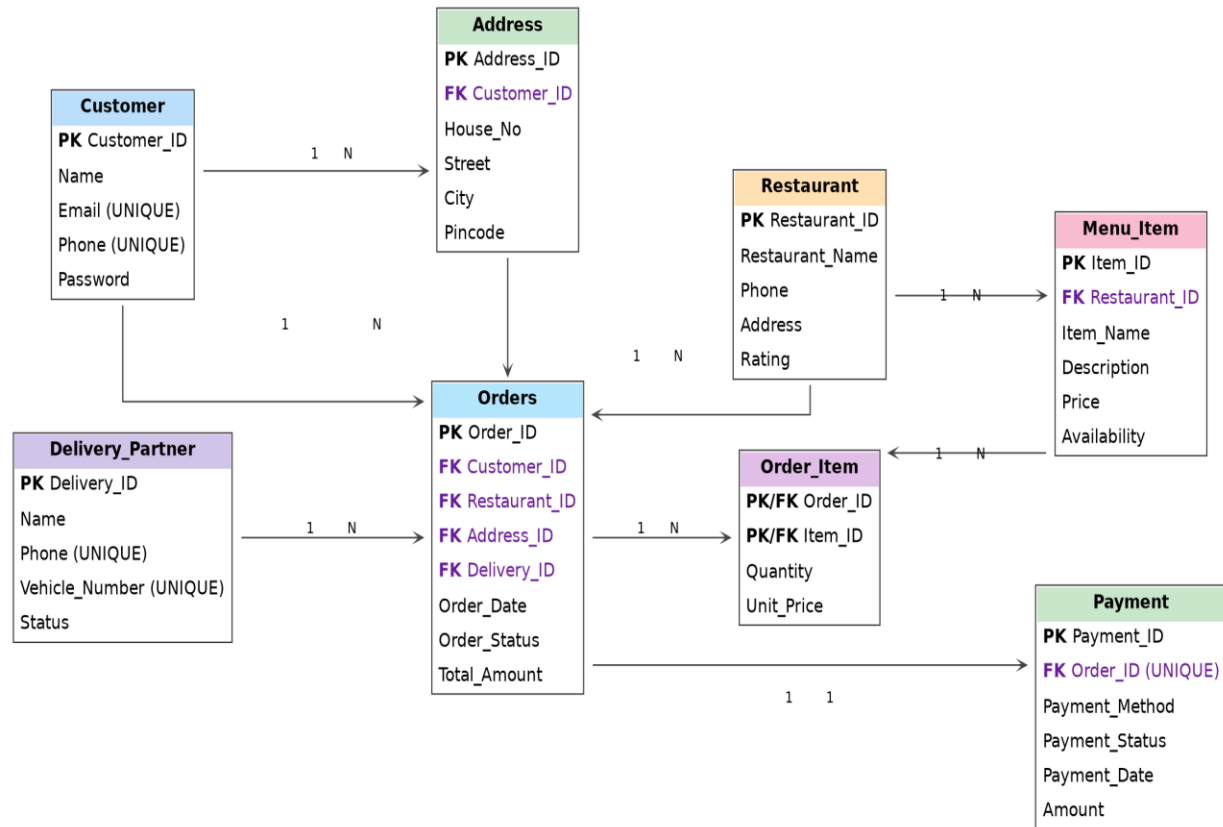
Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique identifier for each payment record
Order_ID	INT	FK, UNIQUE	Order being paid for (references Orders)
Payment_Method	VARCHAR(30)		e.g. UPI, Card, Cash on Delivery
Payment_Status	VARCHAR(30)		e.g. Pending, Successful, Failed, Refunded
Payment_Date	DATETIME		Timestamp the payment was completed
Amount	DECIMAL(10,2)		Amount charged

2. Relationships and Cardinalities

Entity A	Entity B	Cardinality	Description
Customer	Address	1 : N	A customer may save many delivery addresses; each address belongs to one customer.
Customer	Orders	1 : N	A customer may place many orders; each order is placed by exactly one customer.
Restaurant	Menu_Item	1 : N	A restaurant offers many menu items; each menu item belongs to one restaurant.
Restaurant	Orders	1 : N	A restaurant fulfils many orders; each order is placed at one restaurant.
Delivery_Partner	Orders	1 : N	A delivery partner delivers many orders over time; each order is assigned to one delivery partner.
Address	Orders	1 : N	A saved address can be used for many orders; each order ships to exactly one address.
Orders	Menu_Item	M : N	An order can contain many menu items, and a menu item can appear in many orders. Resolved by Order_Item.
Orders	Payment	1 : 1	Each order has exactly one payment record, and each payment record belongs to exactly one order.

3. ER Diagram

The diagram below shows all eight entities, their attributes, primary keys (PK, bold), foreign keys (FK, purple) and the cardinality of every relationship (1, N).



4. Relational Schema

The ER diagram converts directly into the following relations. Primary keys are underlined in bold; foreign keys are shown in purple with their referenced table.

CUSTOMER (Customer_ID **PK**, Name, Email, Phone, Password)

ADDRESS (Address_ID **PK**, Customer_ID **FK** → CUSTOMER.Customer_ID, House_No, Street, City, Pincode)

RESTAURANT (Restaurant_ID **PK**, Restaurant_Name, Phone, Address, Rating)

MENU_ITEM (Item_ID **PK**, Restaurant_ID **FK** → RESTAURANT.Restaurant_ID, Item_Name, Description, Price, Availability)

DELIVERY_PARTNER (Delivery_ID **PK**, Name, Phone, Vehicle_Number, Status)

ORDERS (Order_ID **PK**, Customer_ID **FK** → CUSTOMER, Restaurant_ID **FK** → RESTAURANT, Address_ID **FK** → ADDRESS, Delivery_ID **FK** → DELIVERY_PARTNER, Order_Date, Order_Status, Total_Amount)

ORDER_ITEM (Order_ID **PK, FK** → ORDERS, Item_ID **PK, FK** → MENU_ITEM, Quantity, Unit_Price)

PAYMENT (Payment_ID **PK**, Order_ID **FK, UNIQUE** → ORDERS.Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount)

A complete, executable MySQL script implementing this schema (CREATE DATABASE and all eight CREATE TABLE statements with constraints) is provided as the accompanying file `online_food_delivery_schema.sql`.

5. Primary Key and Foreign Key Identification

Table	Primary Key	Foreign Key(s)
Customer	Customer_ID	—
Address	Address_ID	Customer_ID → Customer.Customer_ID
Restaurant	Restaurant_ID	—
Menu_Item	Item_ID	Restaurant_ID → Restaurant.Restaurant_ID
Delivery_Partner	Delivery_ID	—
Orders	Order_ID	Customer_ID → Customer.Customer_ID; Restaurant_ID → Restaurant.Restaurant_ID; Address_ID → Address.Address_ID; Delivery_ID → Delivery_Partner.Delivery_ID
Order_Item	Order_ID, Item_ID (composite)	Order_ID → Orders.Order_ID; Item_ID → Menu_Item.Item_ID
Payment	Payment_ID	Order_ID → Orders.Order_ID (UNIQUE, enforces 1:1)

6. Relationship Explanation

- Customer – Address (1:N): Each customer can save multiple delivery addresses (home, work, etc.), but every address record belongs to exactly one customer.
- Customer – Orders (1:N): A customer can place many orders over time; each individual order is tied to exactly one customer.
- Restaurant – Menu_Item (1:N): A restaurant maintains a catalog of many menu items; each menu item is offered by exactly one restaurant.
- Restaurant – Orders (1:N): A restaurant fulfils many orders; each order is placed with exactly one restaurant.
- Delivery_Partner – Orders (1:N): A delivery partner can be assigned to and fulfil many orders over time; each order is handled by at most one delivery partner.
- Address – Orders (1:N): A saved address may be reused as the delivery location for several orders; each order specifies exactly one delivery address.
- Orders – Menu_Item (M:N, via Order_Item): One order can include several different menu items, and the same menu item can appear across many different orders. This many-to-many relationship cannot be represented directly in a relational table, so it is resolved with the Order_Item associative table, whose composite primary key (Order_ID, Item_ID) uniquely identifies each line item and also stores Quantity and the Unit_Price locked in at order time.
- Orders – Payment (1:1): Every order generates exactly one payment record, and every payment record is tied to exactly one order. This is enforced by making Payment.Order_ID both a foreign key and UNIQUE.

Design note: Payment.Order_ID carries a UNIQUE constraint to enforce the 1:1 relationship. If the application later needs to support split payments or multiple payment attempts per order, that constraint can be relaxed to a plain foreign key, changing the relationship to 1:N.

7. Final Submission Checklist

- Entities and attributes identified for all 8 tables from Assignment 1.
- Primary keys identified for every entity.
- All relationships and their cardinalities (1:1, 1:N, M:N) documented.

- Digital, clearly labelled ER diagram included (Section 3).
- Relational schema matches the ER diagram exactly (Section 4).
- Foreign keys identified for every table, including the Order_Item associative table resolving the Orders–Menu_Item M:N relationship (Section 5).
- Executable MySQL script (online_food_delivery_schema.sql) provided alongside this document.